<NeuroPulse>

Because you only have 1 head

Steve Hickman, 26-06-22

Problem

Neurological Problems

- WHO: Over 1 in 3 people affected by neurological conditions, the leading cause of illness and disability worldwide
- YaleNews: The massive, hidden burden of neurological disorders

Potential Solution(s)

Neurological Interventions

- PBM Transcranial + Intranasal:
  - Chao LL. Effects of Home Photobiomodulation Treatments on Cognitive and Behavioral Function, Cerebral Perfusion, and Resting-State Functional Connectivity in Patients with Dementia: A Pilot Trial. Photobiomodul Photomed Laser Surg. 2019 Mar;37(3):133-141. doi: 10.1089/photob.2018.4555. : **Key finding:** *8 dementia patients randomized 12 weeks home PBM (Vielight Neuro Gamma, 810nm, 40Hz). PBM group: 4.8-point improvement in ADAS-cog vs 0.8-point decline in usual care. 22.8-point reduction in NPI behavioral symptoms vs 5.3-point worsening in controls. First report of functional connectivity improvements (parietal, DMN) following non-pharmacological intervention. No adverse effects.*
  - Saltmarche AE, Naeser MA, Ho KF, Hamblin MR, Lim L. Significant Improvement in Cognition in Mild to Moderately Severe Dementia Cases Treated with Transcranial Plus Intranasal Photobiomodulation: Case Series Report. Photomed Laser Surg. 2017 Aug;35(8):432-441. doi: 10.1089/pho.2016.4227 : **Key finding:** *5 patients with mild-to-moderate dementia, 12-week home PBM (Vielight Neuro Gamma, 810nm, 40Hz) then 4-week withdrawal. All patients showed significant cognitive improvement on MMSE, ADAS-cog, and NPI-ES at 12 weeks. Gains partially reversed during 4-week withdrawal and recovered when treatment resumed — establishing a dose-response relationship. Foundational Vielight study used by Harvard/BU collaboration.*
  - Zomorrodi R, Loheswaran G, Pushparaj A, Lim L. Pulsed Near Infrared Transcranial and Intranasal Photobiomodulation Significantly Modulates Neural Oscillations: a pilot exploratory study. Sci Rep. 2019;9:6309. doi: 10.1038/s41598-019-42693-x. : **Key finding:** *Single-session 40Hz 810nm NIR PBM (Vielight Neuro Gamma) in a randomized sham-controlled design. Active PBM significantly increased gamma, alpha, and beta EEG power and decreased delta and theta compared to sham — demonstrating for the first time that PBM directly modulates neural oscillations. This established the mechanistic link between 40Hz pulsed PBM and gamma entrainment in humans, justifying NeuroPulse's EEG-adaptive PBM approach.*

- PBM Transcranial:
  - Lee TL, Ding Z, Chan AS. Can transcranial photobiomodulation improve cognitive function? A systematic review of human studies. Ageing Res Rev. 2023 Jan;83:101786. doi: 10.1016/j.arr.2022.101786. : **Key finding:** *Systematic review of human tPBM studies. Majority show cognitive improvement across RCTs and open studies. Key wavelengths 808–850nm most studied. Pulsed (10Hz, 40Hz) outperforms CW. Identifies executive function, memory, and attention as most responsive domains. NeuroPulse relevance: validates 808–830nm wavelength choice and pulsed delivery approach.*
  - Naeser MA, Zafonte R, Krengel MH, et al. Significant improvements in cognitive performance post-transcranial, red/near-infrared light-emitting diode treatments in chronic, mild traumatic brain injury: Open-protocol study. J Neurotrauma. 2014;31(11):1008-1017. doi: 10.1089/neu.2013.3244. : **Key finding:** *11 chronic mTBI patients, 18 LED sessions (870nm + 633nm, 22.2 mW/cm²). Significant improvements in executive function; reduced PTSD symptoms if present. Patients and families reported better social, interpersonal, and occupational functioning. Established transcranial LED PBM for TBI — foundational study for the indication. NeuroPulse relevance: supports TBI/concussion rehabilitation market.*
  - Oliveira BB, Lemos EF, Knob NF, et al. Transcranial photobiomodulation increases cognition and serum BDNF levels in adults over 50 years: A randomized, double-blind, placebo-controlled trial. Neuroscience. 2024; (published online). doi: 10.1016/j.neuroscience.2024.04.001. : **Key finding:** *76 adults over 50 with MCI randomized, 24 sessions whole-head tPBM (RED + NIR LEDs, 2 months). Active group: significantly enhanced cognition AND elevated serum BDNF. First RCT showing dual-wavelength (red + NIR) whole-head tPBM improves cognition AND raises BDNF biomarker. No change in depression or anxiety scores — effect is specific to cognition. Validates NeuroPulse's dual-wavelength (660+810nm) approach.*

Potential Solution(s)

Neurological Interventions (continued)

- PBM Retinal:
  - Shinhmar H, Grewal M, Sivaprasad S, et al. Optically Improved Mitochondrial Function Redeems Aged Human Visual Decline. J Gerontol A Biol Sci Med Sci. 2020;75(9):e49-e52. doi: 10.1093/gerona/glaa155. : **Key finding:** *Randomized controlled study, 670nm light (3 min, once). Single brief exposure significantly improved cone-mediated colour contrast sensitivity in participants aged 37-70 years for one week — restoring thresholds to levels of younger subjects. Demonstrates retinal mitochondrial reactivation via 660-670nm wavelength (NeuroPulse goggle wavelength). UCL Jeffery group — world's leading retinal PBM researchers.*
  - Sivapathasuntharam C, Sivaprasad S, Hogg C, Jeffery G. Aging retinal function is improved by near infrared light (670 nm) that is associated with corrected mitochondrial decline. Neurobiol Aging. 2017 Apr;52:66-70. doi: 10.1016/j.neurobiolaging.2017.01.001.: **Key finding:** *670nm light improves retinal function in aged subjects by correcting mitochondrial decline — enhancing membrane potentials and ATP production and reducing age-related inflammation. Establishes the mitochondrial mechanism specifically in retinal tissue. Since retinal ganglion cells are CNS neurons, this mechanistic finding supports NeuroPulse's visual goggle retinal PBM mode as genuine neuroprotection.*
  - Grewal MK, Sivapathasuntharam C, Chandra S, et al. A Pilot Study Evaluating the Effects of 670 nm Photobiomodulation in Healthy Ageing and Age-Related Macular Degeneration. J Clin Med. 2020;9(4):1001. doi: 10.3390/jcm9041001.: **Key finding:** *31 AMD patients + 11 healthy older adults, daily 670nm light 12 months. In normal ageing (no AMD), significant improvement in scotopic thresholds (p=0.03). No significant change in intermediate AMD — suggesting preventive rather than therapeutic window. Key finding for NeuroPulse: daily retinal NIR in the goggle Mode F is most valuable before AMD develops, supporting the healthy ageing wellness market.*

- 40 Hz Gamma Sensory Stimulation (GENUS / Photic Driving):
  - Iaccarino HF, Singer AC, Martorell AJ, et al. Gamma frequency entrainment attenuates amyloid load and modifies microglia. Nature. 2016;540(7632):230-235. doi: 10.1038/nature20587. : **Key finding:** *Foundational MIT Tsai lab study: 40Hz light flickering reduces amyloid-beta by 40-50% and modifies microglia in visual cortex of Alzheimer's mouse models. Established the entire 40Hz GENUS field. NeuroPulse relevance: the mechanistic justification for all 40Hz photic driving protocols in the gamma clarity preset. Phase III human trial now underway. This Nature paper is cited in virtually all subsequent GENUS research.*
  - He M, Cheng Y, Guo F, et al. Gamma frequency sensory stimulation in mild probable Alzheimer's dementia patients: Results of feasibility and pilot studies. PLoS One. 2022;17(12):e0278412. doi: 10.1371/journal.pone.0278412.: **Key finding:** *Phase 1 (n=43 including MCI and healthy volunteers) + Phase 2A RCT (n=15 mild AD). 40Hz GENUS was safe and induced entrainment in hippocampus, amygdala, insula, and other subcortical structures. After 3 months daily: lesser ventricular dilation and hippocampal atrophy, increased DMN and visual network connectivity, better face-name association delayed recall, improved daily activity rhythmicity. First human RCT demonstrating structural brain preservation with 40Hz sensory stimulation.*

- EEG Neurofeedback:
  - Chiu HJ, Sun CK, Fan HY, et al. Surface electroencephalographic neurofeedback improves sustained attention in ADHD: a meta-analysis of randomized controlled trials. Child Adolesc Psychiatry Ment Health. 2022;16(1):100. doi: 10.1186/s13034-022-00543-1. : **Key finding:** *Meta-analysis of 14 RCTs (n=718), EEG neurofeedback vs comparators for ADHD. Significant improvement in sustained attention (p<0.01). Beta enhancement most effective. Effect maintained through follow-up. NeuroPulse relevance: establishes EEG neurofeedback efficacy for attention — the mechanism behind the NeuroPulse 8-channel neurofeedback system's most prominent consumer claim (Focus).*
  - Chiu HJ, et al. Factors influencing therapeutic effectiveness of electroencephalogram-based neurofeedback against core symptoms of ADHD: a systematic review and meta-analysis. Child Adolesc Psychiatry Ment Health. 2022;16(1):103. doi: 10.1186/s13034-022-00542-2. : **Key finding:** *21 RCTs (n=1261). EEG neurofeedback significantly superior to comparators for inattention, hyperactivity/impulsivity, and global ADHD symptoms from both parent and teacher assessment (all p<0.05). Dose-response relationship: more sessions = better outcomes. Active comparators (not sham) needed for rigorous interpretation. NeuroPulse relevance: largest meta-analysis supporting EEG neurofeedback for ADHD, the most prevalent neurofeedback indication.*

Potential Solution(s)

Neurological Interventions (continued)

- Brainwave Entrainment Stimulation (tACS):
  - Debnath R, Elyamany O, Iffland JR, et al. Theta transcranial alternating current stimulation over the prefrontal cortex enhances theta power and working memory performance. Front Psychiatry. 2025;15:1493675. doi: 10.3389/fpsyt.2024.1493675. : **Key finding:** *28 healthy participants, 5Hz tACS vs sham over left DLPFC. Active tACS significantly improved reaction time in high working memory load trials only. Post-stimulation EEG showed significant increase in 5Hz theta power at stimulation site. Demonstrates DLPFC theta tACS both entrains the target oscillation AND improves the corresponding cognitive function — mechanistic validation of the approach.*
  - Rufener KS, Zaehle T. Systematic review on the modulation of cognitive performance with transcranial alternating current stimulation: Frequency-specific effects. Front Hum Neurosci. 2020;14:600454. doi: 10.3389/fnhum.2020.600454. : **Key finding:** *Systematic review of frequency-specific tACS cognitive effects. Theta-tACS: beneficial for working memory, executive functions, and declarative memory. Gamma-tACS: enhanced auditory and visual perception. Alpha/gamma-tACS: attention improvements (less consistent). Key result for NeuroPulse: frequency specificity is real and predicts cognitive domain — justifying the NeuroPulse preset library's frequency-to-function mapping*.
  - Rakhshan V, Hassani-Abharian P, Joghataei M, et al. Effects of the Alpha, Beta, and Gamma Binaural Beat Brain Stimulation and Short-Term Training on Working Memory: A Within-Subject Randomized Placebo-Controlled Clinical Trial. Biomed Res Int. 2022;2022:8588272. doi: 10.1155/2022/8588272. : **Key finding:** *Systematic review of frequency-specific tACS cognitive effects. Theta-tACS: beneficial for working memory, executive functions, and declarative memory. Gamma-tACS: enhanced auditory and visual perception. Alpha/gamma-tACS: attention improvements (less consistent). Key result for NeuroPulse: frequency specificity is real and predicts cognitive domain — justifying the NeuroPulse preset library's frequency-to-function mapping.*

- Cortical Priming Stimulation (tDCS):
  - Wang J, Yao X, Ji Y, Li H. Cognitive potency and safety of tDCS treatment for major depressive disorder: a systematic review and meta-analysis. Front Hum Neurosci. 2024;18:1458295. doi: 10.3389/fnhum.2024.1458295. : **Key finding:** *Systematic review and meta-analysis, tDCS in MDD. Well-established antidepressant benefits confirmed. Additionally: tDCS improves cognitive function in MDD patients (attention, memory, executive function) beyond symptom reduction. DLPFC anodal tDCS is the standard effective protocol. NeuroPulse relevance: tDCS is effective for both depression AND the cognitive deficits that accompany it — two conditions in the primary target demographic.*
  - Zheng EZ, Wong NML, Yang ASY, Lee TMC. Evaluating the effects of tDCS on depressive and anxiety symptoms from a transdiagnostic perspective: a systematic review and meta-analysis of randomized controlled trials. Transl Psychiatry. 2024;14:289. doi: 10.1038/s41398-024-03003-w. : **Key finding:** *56 RCTs of repeated tDCS sessions with parallel design, transdiagnostic depression and anxiety. tDCS significantly reduced both depressive and anxiety symptoms across diverse clinical conditions. Medium effect size. Findings extend tDCS benefit beyond MDD to anxiety disorders, cancer patients, chronic pain — broadening the indication landscape for NeuroPulse's tDCS module.*
  - Jog MA, Norris V, Pfeiffer P, et al. Personalized High-Definition Transcranial Direct Current Stimulation for the Treatment of Depression. JAMA Netw Open. 2025;8(9):e2531189. doi: 10.1001/jamanetworkopen.2025.31189. : **Key finding:** *71 participants with moderate-severe depression, 12 consecutive days personalized left DLPFC HD-tDCS vs sham. Active HD-tDCS produced statistically significant mood improvement. HD-tDCS provides more focal targeting than conventional tDCS. This trial directly validates NeuroPulse Pro's HD-tDCS pattern capability, and confirms the DLPFC targeting rationale used in NeuroPulse's cortical priming stimulation protocols.*
  - Ji S, Guo Y, Liu Y, et al. Transcranial Direct Current Stimulation Combined With Repetitive Transcranial Magnetic Stimulation for Depression: A Randomized Clinical Trial. JAMA Netw Open. 2024;7(11):e2440888. doi: 10.1001/jamanetworkopen.2024.40888. : **Key finding:** *240 depression patients, 4-arm RCT: tDCS+rTMS vs each alone vs double-sham. Active tDCS + active rTMS produced significantly greater HDRS-24 reduction than either alone or sham after 2 weeks. Combination was more effective with comparable safety. Directly validates NeuroPulse Pro's most important combination protocol: tDCS priming followed by TMS amplifies neuroplasticity beyond either modality alone.*

- Transcranial Magnetic Stimulation (TMS/rTMS):
  - O'Reardon JP, Solvason HB, Janicak PG, et al. Efficacy and safety of transcranial magnetic stimulation in the acute treatment of major depression: a multisite randomized controlled trial. Biol Psychiatry. 2007;62(11):1208-1216. doi: 10.1016/j.biopsych.2007.01.018. : **Key finding:** *301 medication-free patients with treatment-resistant MDD, randomized 10Hz left DLPFC TMS vs sham, 4-6 weeks. Primary endpoint (MADRS) narrowly missed significance (p=0.057) but secondary HDRS measures showed significant improvement. This was the FDA pivotal trial leading to first TMS clearance for depression in 2008. Establishes the predicate for NeuroPulse Pro's TMS module regulatory pathway — cleared for the same indication.*
  - McClintock SM, Reti IM, Carpenter LL, et al. Consensus Recommendations for the Clinical Application of Repetitive Transcranial Magnetic Stimulation (rTMS) in the Treatment of Depression. J Clin Psychiatry. 2018;79(1):16cs10905. doi: 10.4088/JCP.16cs10905. : **Key finding:** *Expert consensus recommendations from the Clinical TMS Society. Response rates: 50-55% with full treatment course; remission 30-35%. rTMS demonstrates efficacy across multiple depression presentations including treatment-resistant cases. Maintenance protocols recommended. NeuroPulse relevance: establishes that TMS Tier 2 clinical market is large (>1M patients eligible in US) and reimbursable (CPT 90867-90869).*

Potential Solution(s)

Neurological Interventions (continued)

- Vagus Nerve Stimulation (Transcutaneous Auricular - taVNS):
  - Redgrave J, Day D, Leung H, et al. Clinical application of transcutaneous auricular vagus nerve stimulation: a scoping review. Disabil Rehabil. 2024;46(25):5730-5760. doi: 10.1080/09638288.2024.2313123. :**Key finding:** *109 studies (2214 active taVNS, 1017 sham taVNS participants) across 21 clinical populations. taVNS shows therapeutic effect across depression, epilepsy, stroke, tinnitus, chronic pain, cognition, and cardiac disorders. Key parameters: 1-4 mA, 20-25 Hz, 200-500μs pulse width most used. NeuroPulse relevance: comprehensive evidence map showing auricular VNS at NeuroPulse's specified parameters (1-25Hz, ≤2mA) is the clinically validated approach across multiple indications.*
  - Shu Y, Zhang J, Huang Y, et al. Transcutaneous auricular vagus nerve stimulation for treatment of drug-resistant epilepsy: a randomized, double-blind clinical trial. Seizure. 2023. doi: 10.1016/j.seizure.2023.01.002. : **Key finding:** *150 drug-resistant epilepsy patients randomized taVNS vs sham, 20 weeks. At 20 weeks: responder rate (>50% seizure reduction) significantly higher in active group. Active group showed significant relative reduction in seizure frequency vs control. Anxiety and depression scales also improved. Largest taVNS RCT in epilepsy. Validates NeuroPulse's auricular VNS safety profile and clinical efficacy.*
  - Austelle CW, O'Leary GH, Thompson S, et al. Accelerated Transcutaneous Auricular Vagus Nerve Stimulation for Inpatient Depression and Anxiety: The iWAVE Open Label Pilot Trial. Neuromodulation. 2025. doi: 10.1016/j.neurom.2025.01.007. *:* **Key finding:** *Open-label pilot, accelerated taVNS for inpatient MDD/anxiety. taVNS modulates DMN functional connectivity in MDD, with connectivity changes correlated with antidepressant response. Establishes that taVNS affects the same neural circuits (DMN) targeted by NeuroPulse's PBM and tACS protocols — validating the combined multi-modal approach targeting convergent circuits.*

- Neural Audio Entrainment (Binaural Beats / Isochronic Tones):
  - Ingendoh RM, Posny ES, Heine A. Binaural beats to entrain the brain? A systematic review of the effects of binaural beat stimulation on brain oscillatory activity, and the implications for psychological research and intervention. PLoS One. 2023;18(5):e0286023. doi: 10.1371/journal.pone.0286023. : **Key finding:** *Systematic review of binaural beat effects on EEG oscillatory activity. Confirms binaural beats can entrain brain oscillations, with theta (4-7Hz) most consistently producing relaxation response and alpha (10Hz) improving mild visuospatial cognition. Effect requires >20 min exposure. Only 60% of participants show measurable entrainment — motivating NeuroPulse's EEG-adaptive closed-loop approach (adjusting to individual responses in real time rather than fixed protocols).*
  - Garcia-Argibay M, Santed MA, Reales JM. Efficacy of binaural auditory beats in cognition, anxiety, and pain perception: A meta-analysis. Psychol Res. 2019;83(2):357-372. doi: 10.1007/s00426-017-0959-2. : **Key finding:** *Meta-analysis of binaural beat effects on cognition, anxiety, and pain. Significant positive effects on anxiety and pain. Moderate effects on attention and memory. Frequency matters: theta for relaxation/anxiety, beta for alertness, gamma for attention and memory. NeuroPulse relevance: confirms audio entrainment as a genuine complement to PBM and BES in multi-modal protocols — not merely adjunctive.*

- HRV Biofeedback (Auricular Clip - Heart Coherence):
  - Pizzoli SFM, Marzorati C, Gatti D, et al. A meta-analysis on heart rate variability biofeedback and depressive symptoms. Sci Rep. 2021;11(1):6650. doi: 10.1038/s41598-021-86149-7. : **Key finding:** *Meta-analysis of 14 RCTs (n=794), HRV biofeedback for depressive symptoms. Random effects analysis: medium mean effect size g=0.38 (95% CI significant). HRV biofeedback reduces depressive symptoms across diverse clinical populations. NeuroPulse relevance: validates the VNS auricular clip's HRV monitoring and biofeedback function as therapeutically meaningful for the depression indication — not just a wellness metric.*
  - Lehrer P, Kaur K, Sharma A, et al. Heart Rate Variability Biofeedback Improves Emotional and Physical Health and Performance: A Systematic Review and Meta Analysis. Appl Psychophysiol Biofeedback. 2020;45(3):109-129. doi: 10.1007/s10484-020-09466-z. : **Key finding:** *Systematic review and meta-analysis (Lehrer, leading HRV biofeedback researcher). Small-to-medium effect sizes for HRV biofeedback on depression, anxiety, anger, and athletic/artistic performance. Paced breathing at resonance frequency (~6 breaths/min) is the optimal technique. Establishes HRV biofeedback as evidence-based across multiple domains. Validates NeuroPulse auricular clip's adaptive breathing guide feature.*

Potential Solution(s)

Neurological Interventions (continued)

- EMDR (Eye Movement Desensitization and Reprocessing):
  - Chen L, Zhang G, Hu M, Liang X. Eye movement desensitization and reprocessing versus cognitive-behavioral therapy for adult posttraumatic stress disorder: systematic review and meta-analysis. J Nerv Ment Dis. 2015;203(6):443-51. doi: 10.1097/NMD.0000000000000306. : **Key finding:** *Meta-analysis of 26 RCTs (1991-2013). EMDR significantly reduced PTSD symptoms (g=-0.662), depression (g=-0.643), anxiety (g=-0.640), and subjective distress (g=-0.956). Duration >60 min per session is optimal. EMDR is WHO-recommended for PTSD. Validates NeuroPulse Pro's EMDR bilateral visual stimulation mode — the device delivers the alternating bilateral stimulation that is the established EMDR mechanism.*
  - Wright SL, Karyotaki E, Cuijpers P, et al. EMDR v. other psychological therapies for PTSD: a systematic review and individual participant data meta-analysis. Psychol Med. 2024;54(8):1580-1588. doi: 10.1017/S0033291723003446. : **Key finding:** *Individual participant data meta-analysis of 8 RCTs (n=346). EMDR significantly reduced PTSD severity, improved treatment response and remission. Comparable to trauma-focused CBT. Identifies moderators of EMDR response. NeuroPulse Pro's EMDR mode delivers the standard bilateral visual stimulation — the most established hardware-supported EMDR component — with the novel addition of real-time EEG monitoring of prefrontal-amygdala connectivity during treatment.*
  - Rodenburg R, Benjamin A, de Roos C, Meijer AM, Stams GJ. Efficacy of EMDR in children: A meta-analysis. Clin Psychol Rev. 2009;29(7):599-606. doi: 10.1016/j.cpr.2009.06.008. [Updated by: NICE 2018 PTSD Guidelines]. : **Key finding:** *EMDR efficacy confirmed in children and adolescents for PTSD. Effect sizes comparable to CBT. WHO recommends EMDR for children and adolescents with PTSD. NeuroPulse relevance: EMDR billing codes (CPT 90837 and H0004) are well-established. NeuroPulse Pro's EMDR goggle + EEG monitoring creates a device-supported EMDR delivery platform suitable for clinic use without requiring physical therapist hand movements.*

- 1064nm Transcranial Photobiomodulation:
  - Yao Y, Huang C, Li Z, Yin H, Huo C, Zhou D, Li X, Wang Y, Tian F, Liu H, Luo Q. "Transcranial photobiomodulation with 1064 nm laser modulates brain electroencephalogram rhythms." Science Advances. 2022;8(8):eabj7390. DOI: 10.1126/sciadv.abj7390. : **Key finding:** Significant improvement in visual working memory (CDA EEG amplitude, p<0.01) in healthy adults (n=20, randomised sham-controlled crossover). Increased cerebral oxygenation (fNIRS) at right prefrontal cortex. Modulation of gamma-band (30–50 Hz) and alpha-band (8–12 Hz) rhythms confirmed. Primary evidence for NeuroPulse 1064nm working memory protocol.
  - (a) Bashkatov AN, et al. "Optical properties of human skin, subcutaneous and mucous tissues in the wavelength range from 400 to 2000 nm." J Phys D Appl Phys. 2005;38(15):2543–2555. DOI: 10.1088/0022-3727/38/15/004. (b) Yaroslavsky AN, et al. "Optical properties of selected native and coagulated human brain tissues in vitro in the visible and near infrared spectral range." Phys Med Biol. 2002;47(12):2059–2073. DOI: 10.1088/0031-9155/47/12/305. (c) Strangman G, et al. "Non-invasive neuroimaging using near-infrared light." Biol Psychiatry. 2002;52(7):679–693. DOI: 10.1016/s0006-3223(02)01550-0. : **Key finding:** At 1064nm, optical absorption coefficient μa is ~3–5× lower than at 660nm in brain tissue; effective attenuation coefficient μeff is lower, resulting in greater effective penetration depth. Modelling predicts 28–35mm effective penetration vs 20–25mm for 808nm and 8–12mm for 660nm. Yaroslavsky et al. (2002) found lowest μa in measurement range for cortical grey matter at 1064nm — directly supports depth-tier claim.
  - Rojas JC, Gonzalez-Lima F. "Neurological and psychological applications of transcranial lasers and LEDs." Biochem Pharmacol. 2013;86(4):447–457. DOI: 10.1016/j.bcp.2013.06.012. : **Key finding:** Cytochrome c oxidase (CCO) absorption spectrum includes band at 1064nm, confirming CCO as primary chromophore for 1064nm PBM and extending established 660/808nm mechanism to longer wavelength. PBM increases CCO enzymatic activity, ATP production, and metabolic rate in neurons. Provides mechanistic grounding for Yao et al. (2022) cortical energy metabolism findings.
  - Hamblin MR. "Shining light on the head: Photobiomodulation for brain disorders." BBA Clin. 2016;6:113–124. DOI: 10.1016/j.bbacli.2016.09.002. : **Key finding:** No adverse events reported in transcranial PBM studies at recommended irradiance levels across reviewed studies. 1064nm within safe window given low tissue absorption and absence of melanin absorption peak. NeuroPulse pulsed operation (25% duty cycle; ~100 mW/cm² average) is at or below parameters reviewed. Thermal and retinal considerations reviewed.

Solution Problem

Most people only have 1 head

- Purchasing tools for each of these modalities adds up
  - VieLight Neuro Pro 2 - $5000
  - Neuronic Light - $1795
  - Vielight Vagus - $699
  - Heartmath Inner Balance - $179
  - BrainMaster Freedom 20R - $24K and up
  - .... some capabilities not yet available to consumers
- Then you have clutter, maintenance (who ya gonna call?), etc.
- No integration or coordination - of protocols or results

You can't use them together because you only have 1 head

Solution Solution

NeuroPulse

Tier	Name	Regulatory	Modalities	Price Range
T1	NeuroPulse Home	FDA-exempt wellness	8	$449–$1,199
T2	NeuroPulse Pro	FDA 510(k) target	11	$4,999–$13,999 + $1,800/yr

Founding design principles:

- Shared platform (one production line, two markets)
- Wired-first USB-C default (zero RF at scalp)
- Autonomous closed-loop EEG-adaptive stimulation without phone (primary competitive moat)
- 5-layer EMF shielding + active Helmholtz cancellation (only consumer brain wearable with measured shielding)
- Modular field-upgradeability via snap-in zone modules
- No mandatory subscription — all core functions offline-capable permanently
- UHDR/SHDR data separation (user health data never accessed by NeuroPulse)

NeuroPulse

CONFIGURATIONS + PRICING

Config	BOM	COGS	Retail	GM%	Modalities included
Core — EEG only	$168–169	$258–260	$449	42%	4-ch EEG · all connectivity · EMF shielding · processor stack · 8GB eMMC
Home Lite	$265–266	$370–372	$599	38%	Core + PBM 5 zones (660+810nm, 600 LEDs) · 8-ch EEG · VNS+HRV clip
Home Standard ★ (flagship)	$405	$540	$849	36%	All T1 modalities (see §3)
Home Premium	$460	$622	$1,199	48%	All T1 + EC lens (+$89 value) · 2yr warranty · priority support
Pro Entry	$833	$1,365	$4,999	73%	All T1 + 21-ch qEEG · 1170nm deep PBM · clinical tACS · HIPAA cloud · sLORETA
Pro Full	$1,506	$2,628	$13,999	81%	All T2 + TMS hub · multi-patient dashboard · scripting API · FHIR R4 · $1,800/yr service

★ **Home Standard box contents:** All T1 modules · hard clamshell case · braided aramid USB-C cable (spare in box) · **45W NeuroPulse branded GaN charger** · S1 opaque shade · interface covers (installed + spare set each type) · mesh cleaning brush · Boa replacement cable + hook tool · moisture-barrier electrode tip hydration caps · humidity indicator card · pre-impregnated cleaning cloth packets

T1 Modalities

PBM Transcranial

- 660–670nm + 808–830nm LEDs (base module)
- **1064nm snap-in smart zone module upgrade (accessory):** Path B smart module architecture — on-module Microchip ATtiny402 I2C slave + 3× Infineon IRLML6344 N-FETs drives 660nm (CH_A), 808nm (CH_B), and 1064nm (CH_C) independently from existing 20-pin FPC connector. 550 LEDs per module (200+200+150). InGaAs PDs (Hamamatsu G12180-010A) for dose metering. Smart module detected via 3.3kΩ ZONE_ID resistor (pin 18, ADC < 1100 counts). Hub enables dedicated LPI2C3 bus (400kHz fast mode) per slot on detection. Hub PCB Rev B adds Vishay DG2788A TIA gain switch per slot (OI-PBM-HW-01, BLOCKING). Base modules unchanged. BOM delta +$23–28, retail $149–199/zone. See NP-FW-PBM1064-001 Rev A, NP-HW-FPC-001 Rev E.
- **T2 combined 1064nm + 1170nm session:** 1064nm smart zone modules (cortical depth) + existing 1170nm laser system (subcortical depth) coordinated by session orchestrator. Three-tier penetration stack: 660nm surface → 1064nm cortical → 1170nm deep. Thermal throttle priority: 1170nm throttled first, then 1064nm CH_C, then CH_B. See NP-FW-PBM1064-001 Rev A §8 and NP-SES-1064-001 Rev A §4.
- Multiple independently addressable zones
- 7 frequency presets: Gamma clarity (40Hz), Alpha calm (10Hz), Theta memory (6Hz), Sleep deep (2Hz), Gamma+theta coupled (40+6Hz split-zone), Focus prime (20Hz), Vascular baseline (CW)

PBM Intranasal

- Bilateral Y-probe · 660nm + 808–830nm per probe

EEG Neurofeedback

- 8-ch semi-dry hydrogel · Fp1/2, F3/4, C3/4, P3/4

BES / tACS (consumer name: Brainwave Entrainment Stimulation)

- 0.5–40Hz · ≤1mA · charge-balanced biphasic

tDCS (consumer name: Cortical Priming Stimulation)

- 0.1–2mA DC · 40μC/cm² hardware limit (safety MCU enforced, app cannot override)

VNS + HRV + HRV Biofeedback

- Auricular clip · auricular branch CN X
- 1–25Hz · ≤2mA · biphasic charge-balanced
- PPG HRV (808–830nm) in same clip
- **HRV Biofeedback Protocol** (software only, no additional hardware):
  - Resonance frequency breathing pacer: default 6 breaths/min (0.1 Hz); personalised to user's peak HRV frequency during first-session sweep (4–7 breaths/min range)
  - Breathing cue delivery: visual ring expanding/contracting in app + optional bone conduction audio cue (uses existing audio hardware)
  - Real-time coherence score: LF peak power / (LF + HF total power), displayed 0–10 colour-coded
  - RMSSD displayed per session; session trend graph over 30 sessions
  - **Four protocols:**
    - Standalone coherence training (5–20 min, breathing pacer + coherence display)
    - HRV + taVNS synchronised: stimulation pulses timed to inspiration phase (PPG R-R interval detects respiratory cycle); optimises noradrenergic modulation window
    - HRV + EEG dual biofeedback: coherence score + EEG band power displayed simultaneously; closed-loop EEG-adaptive frequency adjusts to both signals
    - HRV + PBM: PBM running during HRV coherence training (replicates 2025 multi-modal RCT protocol: PBM + qEEG NF + HRV biofeedback)

Neural Audio Entrainment

- Over-ear planar magnetic 40mm + bone conduction at mastoid
- Binaural beats + isochronic tones + pink/brown noise
- EEG-adaptive frequency (closed-loop)

Visual Stimulation

- IR proximity sensors (940nm) — eye-open detection
- Photoparoxysmal EEG detection at Oz → goggle halt <200ms
- **Mode F (invisible NIR retinal walk):** 808–830nm daily retinal PBM during normal-looking wear
- EMDR L/R alternation · photic driving 0.5–100Hz

Addons & T2 Additional Modalities

Addons

Snap-on shade system:

EC lens (premium, +$89 upgrade / $129 standalone):

T2 Additional Modalities

- **21-ch qEEG wet gel:** Full 10-20 + FC3/FC4 (M1 TMS targeting) + Oz (photoparoxysmal detection) + A1/A2 (linked-ear normative reference, on VNS clips)
- **TMS focal figure-8 coil:** 0.1–0.5T · rTMS + TBS · non-conductive CFRP window at coil site · TMS-gated EMF cancellation (safety MCU gates Helmholtz off 5ms pre-pulse, 50ms post-pulse hold)
- **1170nm deep PBM:** Laser diodes · 35–40mm subcortical depth · TEC stabilisation · ≤1,000 mW/cm²
- **Clinical tACS:** ≤4mA · 16-ch arbitrary waveform
- **sLORETA-guided HD-tDCS:**
- **Cervical VNS (tcVNS) — T2 accessory:**

Clinical Support

- **HIPAA cloud + EHR:** FHIR R4 · multi-patient dashboard · sLORETA source imaging (also drives HD-tDCS targeting) · LSL streaming · scripting API
- **Anonymised session tag:** Random session identifier for clinical multi-patient environments — clinic holds patient-to-tag mapping, NeuroPulse cannot cross-reference